

F70CF Continuous-time Finance - Assignment 2024

Plagiarism

Working in teams is encouraged. It will be useful to bounce ideas off other people and discuss the calculation stage of the process. It can also be beneficial to have someone (not taking the course) proof read your report before you submit it.

However, reports must be written up separately and in your own words. Any ideas, words or coding that you have taken from any other source (including but not limited to: the internet, newspapers, coursework from other students in your class or any other class or cohort) must be properly referenced. You must also acknowledge who you have collaborated with.

Failure to reference work that has been obtained from other sources or to copy the words and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course.

Students must never give hard or soft copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code.

Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.

Full details of the University's plagiarism policy can be found at:
<https://www.hw.ac.uk/students/studies/examinations/plagiarism.htm>

Submission and feedback

The assignment should be submitted via Canvas by **3.30pm** (local time on your campus) on the **15th of November 2024**. The results and feedback will be provided by the end of Week 12 (Friday the 29th of November 2024).

Late Submissions

Coursework that is submitted late (up to a maximum of five working days) will be subject a 30% deduction from the *awarded* mark. For example, if the awarded mark for coursework is 15, and it was submitted late, then the mark for that piece of coursework would be $15 \times 0.7 = 10.5$.

Any coursework submitted more than five days late will be awarded a mark of zero.

It is recognised however that in exceptional circumstances (e.g. illness) a project will legitimately be late. Students so affected should still submit their coursework within five working days of the deadline and should also submit a Mitigating Circumstances application as soon as possible).

Remember to back up your work regularly.

Assignment Exercises & Tasks

The assignment involves coding in R. You must begin your code with `set.seed(1234)`, where you replace 1234 in the line `set.seed(1234)` with the last four digits of your student id. Every time you execute your R code you must execute the whole file including the line `set.seed(...)`.

For Questions 3 and 4, you will need the files `F70CF.R` and `F70CF_assignment_2.R`. Make sure that both files are in your R working directory. You should add your own R code for Questions 3 and 4 to `F70CF_assignment_2.R`.

1. For this question set the number of time units $T = 4$, the number of time points per time unit: $n = 500$ and the number of paths $M = 1,000$.
 - (a) Simulate M paths of a standard Brownian Motion W_t for $t \in [0, T]$ using n time steps per time unit.

[1 mark]

- (b) Use the M paths in Question 1a to calculate an approximate value of the correlations $\text{Corr}(W_2, W_4)$ and $\text{Corr}(W_2, (W_4 - W_3)^3)$.

[1 mark]

- (c) Derive the precise values of $\text{Corr}(W_2, W_4)$ and $\text{Corr}(W_2, (W_4 - W_3)^3)$ and compare them to your answer to Question 1b.

[2 marks]

- (d) Using the simulation from Question 1a, plot a histogram of the random variable $Z := 2W_2 - W_4$.

[1 marks]

- (e) Use the Monte Carlo method to estimate the variance of Z .

[2 marks]

- (f) Provide a rigorous mathematical argument identifying the distribution of Z (by using the properties of Brownian motion discussed in the lectures), and comment on the relation between this theoretical result and your estimation from Question 1e.

[2 marks]

[Total 9 marks]

2. For the simulations in this question, use $T = 4$ and $n = 500$. Consider an SDE given by

$$dZ_t = -\frac{1}{2} \sin(W_t) dt + \cos(W_t) dW_t, \quad (1)$$

where $Z_0 = 0.25$.

- (a) Simulate a path of Z_t for $t \in [0, T]$ with n time steps per time unit using a discrete version of equation (1).

[2 marks]

- (b) Explain how to simulate Z_t in a more precise way.

[2 marks]

- (c) Compare the paths simulated by the methods from Questions 2a and 2b by plotting them in one plot.

[1 mark]

- (d) Explain the reason behind the difference between the two plots.

[2 marks]

[Total 7 marks]

3. In the next two questions, we consider a European Strap option with the following payoff function:

$$X(S_T) = \begin{cases} 2(K - S_T) & \text{if } S_T < K \\ 3(S_T - K) & \text{if } S_T \geq K. \end{cases}$$

Assume the Strap has the time to maturity $T = 4$ years and the strike price is $K = 2$. The underlying security of the Strap is a share with volatility σ (as specified in the F70CF_assignment_2.R file), and no dividends are paid until the maturity date of the Strap.

- (a) Implement the Black-Scholes formula for European put options in R. Comment on your code.

[2 marks]

- (b) Show how a Strap can be replicated with European put and call options. You should present formulae for your answer rather than R code or plots. [2 marks]
- (c) Implement the Black-Scholes formula for a Strap in R using your answers to Questions 3a and 3b. Comment on your code. [2 marks]
- (d) Plot the price, $X_0(S_0)$, of the Strap at time $t = 0$ for a range of share prices $S_0 \in [0, S_{\max}]$, where $S_{\max} = 8$ is defined in variable `S_max` in `F70CF_assignment_2.R`. You should plot the graph for values of S_0 given by `S0 = (0:M) * deltaS` (using the notation in `F70CF_assignment_2.R`. Assume that the interest rate (with continuous compounding) is $r = 0.05$ p.a. - use variable `r`. For your plot use `ylim=c(0,max(BS_x0))` where `BS_x0` is the vector of prices at $t = 0$ for the Strap for different values of S_0 . [2 marks]

[Total 8 marks]

4. In this question you are not allowed to replicate the Strap with put and call options as in Question 3. Instead you must use the finite difference method directly with the payoff of the Strap.
- (a) Use the R function `FD_Euro` defined in `F70CF.R` to find the prices of the Strap for share prices S_0 between 0 and $S_{\max}$. The parameters δS (`deltaS`) and δt (`deltaT`) are given in `F70CF_assignment_2.R`. Use a maximum share price $S_{\max} = 8$ (variable `S_max`). Comment on your code and briefly explain the idea of the finite difference scheme. In particular, comment on the boundary conditions. [3 marks]
- (b) Add the prices for $t = 0$ obtained in Question 4a to the plot created in Question 3d as a line using R's `lines` function. Choose a new colour for this line. [1 mark]
- (c) Comment on the graphs produced in Questions 3d and 4b. [2 marks]

[Total 6 marks]

Please be as succinct as possible in your answers, while still addressing the questions being asked. Your R code should be presented as part of your answer. The code will be marked. Therefore, try to produce clear, readable and efficient code with comments.